Observer-Patch Holography Cosmology Data and Likelihood Contracts

Provenance, Comparability, and Claim Boundaries

B. Müller

October 26, 2026

Paper release: `r2054` **Released:** October 26, 2026

Author affiliation: Bernhard Mueller, Pragma Research Inc.

Abstract

We specify data and likelihood contracts for conditional Observer-Patch Holography (OPH) cosmology. Each comparison binds a finite source, compatible transfer equations, declared datasets and covariance, nuisance parameters, and a common observable projection; the contracts distinguish source-derived quantities, imported inputs and fitted parameters, and account for sample overlap and shared calibration. A retrospective comparison ledger with an independent verifier and machine-checked brackets records where the receipt-conditional values land on Planck, ACT, SPT, BICEP/Keck, DESI and SPARC data. The running, tensor, isocurvature and non-Gaussianity rows are consistent. The registered conditional tilt branch $n_s = 1 - P/48 = 0.96602$, a declared generator density evaluated at the calibrated grain P, lies at the lower 0.612 percent position of the official joint posterior chains of ACT DR6 with Planck, CMB lensing and DESI 2024 (DR1) BAO, whose mean is 0.9743 ± 0.0034, and is consistent with the other combinations; the curvature value is in tension at -2.09σ against DESI DR2 with CMB; the fixed-capacity equation of state is in tension with DESI DR2; and the capacity display is in tension at -2.64 and -2.81σ against the DESI DR2 Λ coordinate while sitting at $+0.62$ and $+0.38$ percent against the Planck central value. Every row is on seen data, so each tension is retrospective under baseline ΛCDM and none is a likelihood ratio.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.
1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.
2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.
3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.
4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.
5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.
6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.
7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

This paper's hypothesis and place in the tower

This paper sits at layer seven as the test interface of the cosmology branches. Its hypothesis is methodological: a cosmological value of the construction counts as evidence only through a contract that binds one finite source, compatible transfer, declared data and covariance, nuisance parameters and a common observable projection, with every measured input that selected a formula declared as an input. It proves no physical statement. It fixes the comparison classes, the required model and data information, and the no-data-use rule, and it records in a deterministic ledger where the receipt-conditional values of the companion papers land on seen data.

No source-derived joint cosmology likelihood exists, because the executable model supplies no covariant source, expanding background, physical scale or radial lift; every ledger row is retrospective and conditional on the premises of its owning paper. A conditional branch is rejected when its frozen protocol fails on the eligible data that the protocol names. Such a failure rejects the tested branch and its premises; it rejects the wider proposal only when those premises are proved necessary.

1 Comparison Classes

Every reported number belongs to one of four classes:

1. **source-derived**: all physical inputs follow from the declared OPH construction without using the comparison data;
2. **imported**: established background or microphysical inputs are declared explicitly;
3. **fit**: one or more parameters are inferred from the same data;
4. **diagnostic**: the calculation tests shape, scale, code behavior, or an interface without supporting a physical OPH claim.

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/main/docs/COMPLETION_PROGRAM.md

The cosmology comparisons in scope are diagnostic or imported. No source-derived joint cosmology likelihood exists. The retrospective comparison ledger of Section 6 classifies every recorded comparison by these classes and by whether the compared value is shared with the standard baseline.

In a likelihood comparison, the event law of the spacetime construction [1], its cone and count-clock hypotheses and their physical calibration count as model information, alongside the background, conserved stress, initial conditions and matter assumptions. A geometric count clock by itself does not define the cosmological transfer or observable likelihood.

2 Required Model Information

A physical comparison must state:

1. the covariant action and source map;
2. the background solution and species content;
3. the gauge-invariant initial conditions;
4. the perturbation, collision, and exchange equations;
5. the observable projection and numerical tolerances;
6. every external input and fitted parameter.

For the modular-charge dark-sector proposal, conserved charge in a supplied comoving volume gives the homogeneous dilution law $\rho_A \propto a^{-3}$. With the separate no-exchange continuity equation it gives a pressureless background. The abundance, relativistic stress, initial perturbations, lensing map, clusters, and nonlinear structure equations are not supplied here.

3 Required Data Information

A comparison must identify the data release, selection, masks, covariance, calibration model, nuisance parameters, priors, and combination rule. Dataset overlap and shared calibration must be included. A diagonal sum is not a joint likelihood when the covariance is non-diagonal.

The model and data layers are separate. Measured values that select a source formula, normalization, branch, or applicability domain are inputs to that comparison rather than OPH outputs.

4 Observable Contracts

Observable family	Required theory output	Required comparison object
Cosmic microwave background (CMB)	C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE}, lensing, foreground model	map or band-power likelihood with covariance and nuisance model
Baryon acoustic oscillations (BAO) and supernovae	$H(z)$, $D_A(z)$, luminosity distance, sound horizon	survey likelihood with calibration covariance
Weak lensing	metric potentials, nonlinear matter power, intrinsic alignments	tomographic likelihood with masks and covariance
Growth and redshift-space distortions (RSD)	$P_m(k,z)$ and $f\sigma_8(z)$	windowed survey likelihood with cross-bin covariance
Clusters	mass function, selection, observable–mass map, lensing calibration	count and calibration likelihood
Galaxies	disk dynamics, gas and stellar nuisances, external environment	galaxy-level likelihood with selection and covariance

5 Diagnostic Results

The reported CMB overlays, compressed background comparisons, and galaxy scaling checks are diagnostic. Their formulas or inputs are selected with knowledge of the comparison data, omit required covariance or nuisance structure, or lack a physical source map. They test numerical paths and conditional formulas and do not establish a cosmological prediction.

Conditional observer spectra and native-source tests. The observer-spectrum calculation propagates a supplied primordial amplitude and shape through a supplied cosmological background and standard temperature/polarization transfer. A conditional sky realization adds cosmic-variance sampling around that mean; it does not select the mean's inputs. Agreement with a measurement-calibrated reference curve, including equality when identical source and transfer inputs are used, verifies a numerical interface and cannot be counted as independent source evidence. Residual root mean squares formed from marginal band-power errors omit inter-bin covariance and nuisance parameters; they are not a likelihood, probability of the theory, or independent fit significance.

The native-repair tests have a different observable and evidence source: absolute stationary load-record variance and inherited/fresh/settled covariance in 61 retained refinement chains. They

constrain those finite mechanisms without using a CMB normalization, but do not identify load as geometric curvature, set a physical background, or supply a primordial tilt. In the executable observer construction, actual native density snapshots on refining source screens have joint Gaussian limits with covariance $\exp(-|\tau - u|A)$ for a positive operator A of spectral dimension two on the screen, with τ a rescaled native attempt count [4, 3]. That construction does not derive the tilt branch $n_s = 1 - P/48$ from its joint records [5]. Their receipts and conditional observer comparisons are archived [7]. An observational test must first connect the native geometric readout to the source, radial and transfer interfaces with their uncertainties.

6 Retrospective Comparison Ledger

A deterministic ledger released with these papers records every comparison of a conditional cosmological value of the construction with public data. Its producer reads the corpus values and the parent receipts, types each public measurement once with its citation, and writes one row per comparison with a class label; an independent verifier recomputes every row. Every row is on seen data. No row is a frozen prediction or a score, and the rule, theory minus measurement over the quoted standard deviation, is fixed before any row is read. Table 1 summarizes the rows.

The rows that agree are, with three exceptions, shared with the standard baseline and support nothing: zero isocurvature, zero non-Gaussianity at the source and zero curvature are the single-field adiabatic expectation, and the fixed-capacity point is the ΛCDM null. The exceptions are the exact linear tensor zero, which requires vanishing tensor initial amplitude and velocity as well as no forcing and whose null test excludes only specified models above the achieved bound, the exact zero running, which discriminates only against running of order 10^{-2}, and the specific tilt value, which the seen combinations place between $+0.7\sigma$ and -2.4σ under the ledger's Gaussian rule. Its only numerical premise is the declared full-collar generator density $P_\star/24$, which no finite source emits, so the tilt has the epistemic status of the capacity display: a declared one-parameter map evaluated at a pixel located from the measured fine-structure constant. Its distance from the data discriminates that declared map, not a source derivation. In the official ordered ACT DR6, Planck and DESI joint chains, 0.612 percent of the posterior mass lies at or below the branch, against about 0.74 percent for the Gaussian summary; the diagnostic clock alternative 0.96484 lies at 0.200 percent [2, 5]. Absence of an emitted tensor source alone does not establish the tensor-zero premise; source-free homogeneous modes are allowed. Slow roll supplies no universal positive tensor lower bound, so a finite upper limit cannot distinguish the entire model class from zero. The tensions are the fixed-capacity equation of state, at 2.6 to 3.9σ across the DESI DR2 combinations under the moment diagnostic and with monotone-branch posterior fractions between 0.007 and 0.11 percent, the curvature value from DESI DR2 with CMB at 2.1σ, and the capacity display against the DESI DR2 Λ coordinate at -2.64 and -2.81σ, while the same candidates sit at $+0.62$ and $+0.38$ percent against the Planck-central value. These are retrospective diagnostics of receipt-conditional values. FZ-01, frozen on 2026-07-17 with a Bitcoin attestation, registers four cosmology targets: the capacity band $N \in [3.30836, 3.31137] \times 10^{122}$, the tilt menu $\{0.9660214956, 0.96484114303\}$, the acceleration readings 1.184737388 and $1.179018696 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, and the point $(w_0, w_a) = (-1, 0)$. Every comparison in Table 1 uses data published before that freeze, so each is a postdiction, and no FZ-01 verdict is registered; neither closure candidate lies in the registered capacity band [6]. The tilt FZ-15 and fixed-capacity FZ-13 protocols were frozen on 2026-10-04, with register status `frozen_stamped_upgrade_pending`; their complete protocols and anchoring requirements govern eligible comparisons. The registered FZ-15 comparison is the first eligible Simons Observatory

LAT primary-CMB release, which the ACT DR6 reanalysis leaves unchanged. The tensor and running/isocurvature texts FZ-16 and FZ-17 are unadopted proposals and define no comparison. A ledger row alone does not define a score: the combination, statistic and thresholds must be fixed before the eligible release under the operative registration.

7 No-Data-Use Boundary

A source-derived calculation must expose its dependency graph. The source side may use mathematical constants, declared OPH finite artifacts, and explicitly imported physical inputs. It may not use the observed quantity being claimed as an output, a fit to that quantity, or a proxy calibrated from the same dataset.

This provenance rule is enforced through the declared dependency graph and data-use classification.

8 Theorem scope

The screen-spectrum and radial-lift mathematics are conditional theorems. One-shell inversion is nonunique; physical source dilation and radial cross-covariance tomography are two established sufficient uniqueness routes. Neither route is supplied by the native screen energy, which is not boundedly congruent to the static boundary action of a smooth compact three-dimensional Dirichlet bulk; a radial lift needs a radial observation algebra as an added premise [3]. Finite source instantiation, repair-charge cosmological source, Boltzmann bridge, and joint likelihood are not supplied. All cosmological data products are diagnostic. No cosmological result carries an OPH falsification verdict.

References

[1] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf.

[2] T. Louis et al. (2025). The Atacama Cosmology Telescope: DR6 power spectra, likelihoods and ΛCDM parameters. arXiv:2503.14452. https://arxiv.org/abs/2503.14452.

[3] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/model/physical_space_time/OCCUPATION_FIELD.md.

[4] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/model/observer_time_fields/REFINEMENT.md.

[5] FloatingPragma (2026). OPH predictions and public-data comparisons. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/docs/research/PUBLIC_DATA_PREDICTION_AUDIT.md.

[6] FloatingPragma (2026). Frozen prediction ladder. https://github.com/FloatingPragma/observer-patch-holography/blob/r2054/docs/FROZEN_PREDICTION_LADDER.md.

[7] FloatingPragma (2026). Observer-dynamics analysis archive: native-repair receipts and conditional observer comparisons. https://github.com/FloatingPragma/observer-patch-holography/tree/r2054/evidence/observer_dynamics_20260925/sim-analysis/codex.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.

Quantity	Value	Public measurement and distance	Class
n_s	$1 - P_\star/48 = 0.96602$	Planck 2018 0.9649 ± 0.0042 ($+0.27\sigma$); SPT-3G with Planck 0.9636 ± 0.0035 ($+0.69\sigma$); SPT, Planck and ACT 0.9684 ± 0.0030 (-0.79σ); ACT DR6 with Planck 0.9709 ± 0.0038 (-1.28σ); with lensing and DESI 2024 (DR1) BAO 0.9743 ± 0.0034: lower 0.612 percent posterior position in the official chains (Gaussian summary -2.43σ)	declared-density value at the calibrated grain, like the capacity display
$\mathrm{d}n_s/\mathrm{d}\ln k$	0	Planck 2018 -0.0045 ± 0.0067 ($+0.67\sigma$); ACT DR6 with Planck, lensing and BAO 0.0062 ± 0.0052 (-1.19σ)	conditional theorem; discriminates only against running of order 10^{-2}
r	0	BICEP/Keck 2018 $r_{0.05} < 0.036$; the fourth Planck release with BK18 and BAO $r < 0.032$; consistent	requires zero tensor initial data and forcing; tests specified alternatives
β_{iso}	0	Planck 2018 uncorrelated CDI $100\beta_{\mathrm{iso}} < 3.5,\ 3.8,\ 3.9$ at the three reference scales; consistent	shared baseline
f_{NL}	0 at the source	Planck 2018 local -0.9 ± 5.1 ($+0.18\sigma$), equilateral -26 ± 47, orthogonal -38 ± 24	shared baseline
Ω_K	0	Planck 2018 with lensing -0.0106 ± 0.0065 ($+1.63\sigma$); with BAO 0.0007 ± 0.0019 (-0.37σ); DESI DR2 with CMB 0.0023 ± 0.0011 (-2.09σ)	shared baseline
(w_0, w_a)	$(-1, 0)$	DESI DR2 BAO with CMB 2.61σ; with Pantheon+ 2.60σ; with Union3 3.38σ; with DESY5 3.93σ (Gaussian moment diagnostic of the official chains)	conditional theorem; tension
$\Lambda\ell_P^2$	$3\pi/N$, $N = 3.29$ and 3.30×10^{122}	Planck-central Λ: $+0.62$ and $+0.38$ percent; DESI DR2 with CMB display $(2.968 \pm 0.040) \times 10^{-122}$: -3.5 and -3.8 percent (-2.6 and -2.8σ)	closure-candidate display
a_0	no source-derived value; FZ-01 readings 1.1847 and $1.1790 \times 10^{-10}\,\mathrm{m\,s^{-2}}$	SPARC calibration $1.161 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at mass-to-light 0.5; ratio to $c^2\sqrt{\Lambda/3}$ equal to 0.214	fitted comparison value

Table 1: Rows of the cosmology postdiction ledger. Distances are theory minus measurement in units of the quoted standard deviation; bounds are consistent when the value lies below the bound. All data are seen. SPARC values are ineligible for scoring the FZ-01 acceleration readings.